

Natural language statement

Domain

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3.6.8. Definition. Let $F : X \rightarrow Y$ be a mapping between measure spaces $(X, \mathcal{A}, \mu)$ and $(Y, \mathcal{B}, \nu)$. We shall say that F has Lusin's property (N) with respect to the pair (μ, ν) if $\nu(F(A)) = 0$ for every set A in $\mathcal{A}$ with $\mu(A) = 0$. 3.6.9. Theorem. Let $F : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be a Lebesgue measurable mapping. Then F has Lusin's property (N) with respect to Lebesgue measure precisely when F takes all Lebesgue measurable sets to Lebesgue measurable sets.

Measure theory

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def HasLusinPropertyN {X Y : Type*} [MeasurableSpace X] [MeasurableSpace Y]
  (F : X → Y) (μ : Measure X) (ν : Measure Y) : Prop :=
  forall A : Set X, NullMeasurableSet A μ => μ A = 0 => ν (F '' A) = 0

theorem hasLusinPropertyN_iff_maps_nullMeasurableSet
  {n : ℕ} {F : (Fin n → ℝ) → (Fin n → ℝ)} (hF : NullMeasurable F
volume) :
  HasLusinPropertyN F volume volume <=>
    forall A : Set (Fin n → ℝ),
      NullMeasurableSet A volume => NullMeasurableSet (F '' A) volume :=
  by
    sorry
```

8.6.2. Theorem. Let X be a complete separable metric space and let $\mathcal{M}$ be a family of Borel measures on X . Then the following conditions are equivalent: (i) every sequence $\{\mu_n\}$ subset $\mathcal{M}$ contains a weakly convergent subsequence; (ii) the family $\mathcal{M}$ is uniformly tight and uniformly bounded in the variation norm. The above conditions are equivalent for any complete metric space X if $\mathcal{M} \subset \mathcal{M}_t(X)$.

Measure theory

```
noncomputable def signedMeasureIntegral {X : Type*} [TopologicalSpace X]
[MeasurableSpace X]
  (s : SignedMeasure X) (f : X → ℝ) : ℝ :=
  (integral x, f x ds.toJordanDecomposition.posPart) -
  integral x, f x ds.toJordanDecomposition.negPart

def weakly_converges_signed {X : Type*} [TopologicalSpace X]
[MeasurableSpace X]
  (s : ℕ → SignedMeasure X) (t : SignedMeasure X) : Prop :=
  forall f : X → ℝ,
    Tendsto (fun n => signedMeasureIntegral (s n) f) atTop (nhds
      (signedMeasureIntegral t f))

def relatively_sequentially_weakly_compact_signed {X : Type*}
[TopologicalSpace X]
[MeasurableSpace X] (S : Set (SignedMeasure X)) : Prop :=
  forall s : ℕ → SignedMeasure X, (forall n, s n in S) =>
    exists phi : ℕ → ℕ, StrictMono phi /\ exists t,
      weakly_converges_signed (fun n => s (phi n)) t

def UniformlyBoundedInTotalVariation {Omega : Type*} [MeasurableSpace Omega]
(S : Set (SignedMeasure Omega)) : Prop :=
  exists C : ℝ, C >= 0, forall s in S, s.totalVariation univ <= C

theorem bogachev_8_6_2_prokhorov_signed_measures
  {X : Type*} [MetricSpace X] [CompleteSpace X]
[MeasurableSpace X] [BorelSpace X] (S : Set (SignedMeasure X)) :
  (SecondCountableTopology X =>
    (relatively_sequentially_weakly_compact_signed S <=>
      IsTightMeasureSet ((fun s : SignedMeasure X => s.totalVariation) ''
        S) /\
        UniformlyBoundedInTotalVariation S)) /\
    ((forall s in S, IsTightMeasureSet {s.totalVariation}) =>
      (relatively_sequentially_weakly_compact_signed S <=>
        IsTightMeasureSet ((fun s : SignedMeasure X => s.totalVariation)
          '' S) /\
          UniformlyBoundedInTotalVariation S)) := by
  sorry
```

5.5.4. Proposition. Let f be a function on the real line and let E be a measurable set such that at every point of E the function f is differentiable. Then $\lambda(f(E)) \leq \int_E |f'(x)| dx$. In particular, the function f on E has Lusin's property (N). If for all x in E we have $|f'(x)| \leq L$, then $\lambda(f(E)) \leq L \lambda(E)$.

Analysis

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def HasLusinPropertyN0n {X Y : Type*} [MeasurableSpace X] [MeasurableSpace
Y]
  (F : X → Y) (E : Set X) (μ : Measure X) (ν : Measure Y) : Prop :=
  forall A : Set X, NullMeasurableSet A μ => A subset E => μ A = 0 => ν
(F '' A) = 0

theorem proposition_5_5_4
  (f : ℝ → ℝ) (E : Set ℝ) (hE : NullMeasurableSet E volume)
(hf : forall x in E, DifferentiableAt ℝ f x) :
  (volume (f '' E) <= integral^~ x in E, ENNReal.ofReal (abs (deriv f x))
    dvolume) /\
    HasLusinPropertyN0n f E volume volume /\
    forall L : ℝ,
      (forall x in E, abs (deriv f x) <= L) =>
        volume (f '' E) <= ENNReal.ofReal L * volume E := by
  sorry
```

3.7.1. Theorem. If the mapping F is injective on U , then, for any measurable set A subset U and any Borel function g in

Analysis

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theorem bogachev_3_7_1_change_of_variables_in_ℝn
  {n : ℕ} {U A : Set (Fin n → ℝ)} {F : (Fin n → ℝ) → (Fin n → ℝ)}
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$L^1(\mathbb{R}^n)$, one has the equality $\int_A g(F(x)) J_F(x) dx = \int_{F(A)} g(y) dy$.		<pre> {g : (Fin n -> R) -> R} (hU : IsOpen U) (hF : ContDiffOn R 1 F U) (hinj : InjOn F U) (hA : NullMeasurableSet A volume) (hAU : A subset U) (hg : Integrable g volume) : (integral x in A, g (F x) * (fderivWithin R F U x).det dvolume) = integral y in F '' A, g y dvolume := by sorry </pre>
4.5.9. Theorem. Let μ be a finite nonnegative measure. A family F of μ-integrable functions is uniformly integrable if and only if there exists a nonnegative increasing function G on $[0, +\infty)$ such that $\lim_{t \rightarrow +\infty} G(t)/t = \infty$ and $\sup_{f \in F} \int G(f(x)) \mu(dx) < \infty$. In such a case, one can choose a convex increasing function G.	Measure theory	<pre> theorem bogachev_4.5.9_de_la_vallee_poussin {Omega : Type*} [MeasurableSpace Omega] {mu : Measure Omega} [IsFiniteMeasure mu] {F : Iota -> Omega -> R} (hF : forall i : Iota, Integrable (F i) mu) : let superlinear := fun G : R -> R => (forall t : R, 0 <= t -> 0 <= G t) /\ MonotoneOn G (Ici (0 : R)) /\ Tendsto (fun t : R => G t / t) atTop atTop (UniformIntegrable F 1 mu <=> exists G : R -> R, superlinear G /\ exists C : R>=0, forall i : Iota, integral^~ x, ENNReal.ofReal (G F i x) dmu <= C) /\ (UniformIntegrable F 1 mu -> exists G : R -> R, superlinear G /\ ConvexOn R (Ici (0 : R)) G /\ exists C : R>=0, forall i : Iota, integral^~ x, ENNReal.ofReal (G F i x) dmu <= C) := by sorry </pre>
4.6.3. Theorem. Let a sequence of measures μ_n in the space $M(X, A)$ be such that $\lim_{n \rightarrow \infty} \mu_n(A)$ exists and is finite for every set A in A. Then: (i) the formula $\mu(A) = \lim_{n \rightarrow \infty} \mu_n(A)$ defines a measure μ in $M(X, A)$; (ii) there exist a nonnegative measure ν in $M(X, A)$ and a bounded nondecreasing nonnegative function α on $[0, +\infty)$ such that $\lim_{n \rightarrow \infty} \alpha(t) = 0$ and $\sup_n \mu_n(A) \leq \alpha(\nu(A))$, for all A in A. In particular, $\sup_n \mu_n < \infty$ and the sequence $\{\mu_n\}$ is uniformly countably additive; (iii) if a nonnegative measure λ in $M(X, A)$ is such that $\mu_n \ll \lambda$ for all n, then $\lim_{n \rightarrow \infty} \sup \{\mu_n(A) : A \in A, \lambda(A) \leq t, n \in \mathbb{N}\} = 0$. 4.6.2. Definition. Let M be a family of real measures on a sigma-algebra A. This family is called uniformly countably additive if, for every sequence of pairwise disjoint sets A_i, the series $\sum_{i=1}^{\infty} \mu(A_i)$ converges uniformly in μ in M, i.e., for every $\epsilon > 0$, there exists n_ϵ such that $\sum_{i=1}^n \mu(A_i) < \epsilon$ for all $n \geq n_\epsilon$ and all μ in M.	Measure theory	<pre> def UniformlyBoundedInTotalVariation {Omega : Type*} [MeasurableSpace Omega] (S : Set (SignedMeasure Omega)) : Prop := exists C : R>=0, forall s in S, s.totalVariation univ <= C def UniformlyCountablyAdditive {Omega : Type*} [MeasurableSpace Omega] (S : Set (SignedMeasure Omega)) : Prop := forall A : N -> Set Omega, (forall n : N, MeasurableSet (A n)) -> (forall i j : N, i != j -> Disjoint (A i) (A j)) -> forall eps : R, 0 < eps -> exists N : N, forall s in S, forall n >= N, s (union k : {k // n <= k}, A k.1) < eps def UniformlyAbsolutelyContinuous {Omega : Type*} [MeasurableSpace Omega] (S : Set (SignedMeasure Omega)) (mu : Measure Omega) : Prop := forall eps : R, 0 < eps -> exists delta : R, 0 < delta /\ forall s in S, forall A : Set Omega, MeasurableSet A -> mu A <= ENNReal.ofReal delta -> s A < eps theorem bogachev_4.6.3_nikodym_vitali_hahn_saks {Omega : Type*} [MeasurableSpace Omega] {s : N -> SignedMeasure Omega} (hlim : forall A : Set Omega, MeasurableSet A -> exists l : R, Tendsto (fun n : N => s n A) atTop (nhds l)) : exists sLim : SignedMeasure Omega, (forall A : Set Omega, MeasurableSet A -> Tendsto (fun n : N => s n A) atTop (nhds (sLim A))) /\ (exists nu : FiniteMeasure Omega, exists alpha : R>=0 -> R>=0, Monotone alpha /\ (exists C : R>=0, forall t : R>=0, alpha t <= C) /\ Tendsto alpha (nhds 0) (nhds 0) /\ forall n : N, forall A : Set Omega, MeasurableSet A -> s n A <= (alpha (nu A) : R)) /\ UniformlyBoundedInTotalVariation (range s) /\ UniformlyCountablyAdditive (range s) /\ forall lam : FiniteMeasure Omega, (forall n : N, s n <= v (lam : Measure Omega).toENNRealVectorMeasure) -> UniformlyAbsolutelyContinuous (range s) (lam : Measure Omega) := by sorry </pre>
9.12.37. Corollary. Let $\mu_1, \dots, \mu_n$ be atomless Borel probability measures on a Souslin space X. Then, for every Borel probability measure ν on X, there exists a Borel transformation $T : X \rightarrow X$ such that $\mu_i \circ T^{-1} = \nu$ for all i.	Measure theory	<pre> def is_atomless_measure {X : Type*} [MeasurableSpace X] (mu : Measure X) : Prop := forall A : Set X, MeasurableSet A -> 0 < mu A -> exists B : Set X, MeasurableSet B /\ B subset A /\ 0 < mu B /\ mu B < </pre>

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$\leq n$. 7.14.15. Definition. Let $(M, \mathcal{M}, \mu)$ be a space with a nonnegative measure. An element A subset M is called an atom of the measure μ if $\mu(A) > 0$ and every element B in M that is contained in A has measure either zero or $\mu(A)$. A measure without atoms is called atomless.		μA theorem bogachev_9_12_37_simultaneous_transport <pre>{X : Type*} [MeasurableSpace X] [TopologicalSpace X] [T2Space X] [SouslinSpace X] [BorelSpace X] {n : ℕ} (mu : Fin n → Measure X) [forall i, IsProbabilityMeasure (mu i)] (nu : Measure X) [IsProbabilityMeasure nu] (hmu : forall i, is_atomless_measure (mu i)) : exists T : X → X, forall i, MeasurePreserving T (mu i) nu := by sorry</pre>
10.5.4. Theorem. For every complete probability measure μ, there exists a lifting on $\text{Lin}(\mu)$. 10.5.1. Definition. Let $(X, \mathcal{A}, \mu)$ be a measurable space with a nonnegative measure μ (possibly with values in $[0, +\infty]$) and let $\text{Lin} \mathcal{A}$ be the space of all bounded $\mathcal{A}$-measurable functions. A lifting on $\text{Lin} \mathcal{A}$ is a mapping $L : \text{Lin} \mathcal{A} \rightarrow \text{Lin} \mathcal{A}$ satisfying the following conditions: (i) $L(f) = f$ μ-a.e.; (ii) $L(f)(x) = L(g)(x)$ for all x in X if $f = g$ μ-a.e.; (iii) $L(f)(x) = 1$ for all x in X if $f = 1$ μ-a.e.; (iv) $L(\alpha f + \beta g)(x) = \alpha L(f)(x) + \beta L(g)(x)$ for all x in X, f, g in $\text{Lin} \mathcal{A}$ and α, β in $\mathbb{R}$; (v) $L(fg)(x) = L(f)(x)L(g)(x)$ for all x in X, f, g in $\text{Lin} \mathcal{A}$.	Measure theory	<pre>structure LInfinityLifting {X : Type*} [MeasurableSpace X] (mu : Measure X) where toFun : (X → ℝ) → X → ℝ maps_bounded_measurable : forall f, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> Measurable (toFun f) /\ exists C : ℝ, forall x, toFun f x <= C representative : forall f, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> toFun f = ^m[mu] f congr_ae : forall f g, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> (Measurable g /\ exists C : ℝ, forall x, g x <= C) -> f = ^m[mu] g -> toFun f = toFun g map_ae_one : forall f, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> f = ^m[mu] (1 : X → ℝ) -> toFun f = (1 : X → ℝ) map_add : forall f g, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> (Measurable g /\ exists C : ℝ, forall x, g x <= C) -> toFun (f + g) = toFun f + toFun g map_smul : forall c : ℝ, forall f, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> toFun (c * f) = c * toFun f map_mul : forall f g, (Measurable f /\ exists C : ℝ, forall x, f x <= C) -> (Measurable g /\ exists C : ℝ, forall x, g x <= C) -> toFun (f * g) = toFun f * toFun g theorem bogachev_10_5_4_lifting {X : Type*} [MeasurableSpace X] (mu : Measure X) [IsProbabilityMeasure mu] [mu.IsComplete] : Nonempty (LInfinityLifting mu) := by sorry</pre>
9.1.9. Theorem. Let f be a mapping from a topological space X to a topological space Y with a Radon measure ν. Suppose that there exists an increasing sequence of compact sets K_n subset X such that f is continuous on every K_n and $\lim_{n \rightarrow \infty} \nu(K_n) = \ \nu\$. Then, there exists a Radon measure μ on X with $\mu _{f^{-1}K_n} = \nu _{K_n}$. In addition, this measure can be chosen with the property $\ \mu\ = \ \nu\$. In particular, this is true if X and Y are compact and f is a continuous surjection.	Measure theory	theorem bogachev_9_1_9_radon_preimage_from_compact_approximation <pre>{X Y : Type*} [TopologicalSpace X] [MeasurableSpace X] [BorelSpace X] [TopologicalSpace Y] [MeasurableSpace Y] [BorelSpace Y] {f : X → Y} {nu : SignedMeasure Y} (hnu : Measure.InnerRegular nu.totalVariation) (K : ℕ → Set X) : (Monotone K -> (forall n, IsCompact (K n)) -> (forall n, ContinuousOn f (K n)) -> Tendsto (fun n => nu.totalVariation (f '' K n)) atTop (nhds (nu.totalVariation univ)) -> exists mu : SignedMeasure X, Measure.InnerRegular mu.totalVariation /\ mu.totalVariation univ = nu.totalVariation univ /\ forall A : Set Y, MeasurableSet A -> exists B : Set X, MeasurableSet B /\ (forall^m x dmu.totalVariation, x in B <=> x in f ^-1' A) /\ mu B = nu A) /\ (CompactSpace X -> CompactSpace Y -> Continuous f -> Function.Surjective f -> exists mu : SignedMeasure X, Measure.InnerRegular mu.totalVariation /\ mu.totalVariation univ = nu.totalVariation univ /\ forall A : Set Y, MeasurableSet A -> exists B : Set X, MeasurableSet B /\ (forall^m x dmu.totalVariation, x in B <=> x in f ^-1' A) /\ mu B = nu A) := by sorry</pre>

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4.7.75. (G. Hardy) Let f in $L_p(0, +\infty)$, where $p > 1$. Show that the functions $\phi(x) = x^{p-1} \int_0^x f(t) dt$ and $\psi(x) = \int_0^x t^{p-1} f(t) dt$ belong to $L_p(0, +\infty)$ as well.	Analysis	<pre>theorem hardy_average_and_tail_memLp (f : ℝ → ℝ) (p : ℝ) (hp : 1 < p) (hf : MemLp f (ENNReal.ofReal p) (volume.restrict (Ioi (0 : ℝ)))) : MemLp (fun x : ℝ => (1 / x) * (integral t in (0 : ℝ)..x, f t dvolume)) (ENNReal.ofReal p) (volume.restrict (Ioi (0 : ℝ))) /\ MemLp (fun x : ℝ => integral t in Ioi x, f t / t dvolume) (ENNReal.ofReal p) (volume.restrict (Ioi (0 : ℝ))) := by sorry</pre>